Can Computational Graph Structure Distinguish Surface Recall from Explanatory Reasoning in Language Models?

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Abstract

We present two complementary findings on the computational distinction between surface-level factual recall and explanatory reasoning in language models.

First, a negative result: Five iterations of confound elimination — controlling for word frequency, readability metrics, N-gram overlap, and lexical diversity across three model scales (GPT-2 Small, Gemma-2 2B, Mistral 7B) — fail to reliably separate matched prompt pairs designed to elicit each response type. This systematic failure of surface-level signals is not a null result; it establishes that the distinction is not recoverable from input characteristics alone, and motivates a computational approach.

Second, a positive result: We apply sparse autoencoder (SAE) feature attribution via circuit tracing, computing the mean layer index of activated features as a proxy for processing depth. Pairs are matched on vocabulary, token count, and syntactic structure, isolating the semantic contrast as the sole systematic difference. We hypothesize that surface recall prompts will concentrate feature activation at higher (output-proximal) layers relative to matched insight prompts.

An exploratory phase (30 pairs submitted to Gemma-2-2B; 2 valid after CUDA attribution failures) showed both valid pairs in the predicted direction (mean delta = +0.326). We then pre-registered the directional hypothesis and replicated on 15 held-out pairs (n=11 valid; 4 excluded for attribution extraction failure), obtaining mean delta = +0.118 (SD = 0.192), $t(10) = 2.04$, $p = 0.035$ (one-tailed paired t-test, pre-registered). This result is directionally consistent but fragile: it does not survive Bonferroni correction for multiple samples, leave-one-out sensitivity analysis shows p exceeds 0.05 for 7 of 11 subsets, and the nonparametric Wilcoxon test is marginal ($p = 0.051$).

In a second pre-registered replication on 30 new held-out pairs — no topic overlap with prior 45, same matching criteria — all 30 yield valid bilateral attributions (zero exclusions).

Results: mean delta = +0.257 (SD = 0.313), t(29) = 4.50, p = 0.000051 (one-tailed), Cohen's d = 0.82, 83.3% of pairs in the predicted direction. The result is robust to Bonferroni correction, Wilcoxon test (p = 0.000055), leave-one-out analysis (all 30 subsets p < 0.001), and control for a feature-count asymmetry confound (r = −0.13 between asymmetry and delta, p = 0.50). The effect size matches the exploratory estimate (0.257 vs. 0.326), indicating the n=11 result was underpowered rather than inflated.

These results establish that feature activation depth in sparse circuit representations captures a meaningful computational distinction invisible to surface text analysis. Generalization to larger models and naturalistic uncontrolled text remains the immediate direction for future work.

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1. Introduction

A persistent challenge in language model interpretability is distinguishing *what* a model knows from *how deeply it reasons*. A model that correctly states "the mitochondria is the powerhouse of the cell" and one that can explain oxidative phosphorylation in causal terms may produce similarly fluent output — but the underlying computation differs. Characterizing that difference has implications for interpretability, evaluation, and understanding the limits of language model cognition.

One natural hypothesis is that this distinction is recoverable from input characteristics. Surface recall prompts — questions requiring retrieval of stored facts — might differ measurably from insight prompts — questions requiring causal or mechanistic reasoning — in their lexical or syntactic properties. A simpler vocabulary, shorter sentences, more common N-grams: these might all correlate with surface processing.

We test this hypothesis rigorously and find it fails. Across five rounds of confound elimination and increasingly precise statistical controls at three model scales, surface text statistics do not separate carefully matched prompt pairs. This negative result is substantive: it means evaluating reasoning depth from surface text alone is fundamentally limited.

We then ask whether the distinction is recoverable from the model's internal computation. Using sparse autoencoder (SAE) feature attribution via circuit tracing, we measure where in the network the model concentrates its processing for each prompt type. The core hypothesis: if surface recall is direct lookup and insight reasoning is distributed computation, surface prompts should activate features closer to the output layers, while insight prompts should engage earlier, more distributed representations.

This hypothesis is supported in a pre-registered power replication (n=30, p = 0.000051,

Cohen's d = 0.82), survives multiple robustness checks including nonparametric tests and confound controls, and is consistent across all three independent samples. An initial replication at smaller sample size (n=11) is directionally consistent but statistically fragile, underscoring the importance of adequate power for this effect size.

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2. Related Work

2.1 Mechanistic Interpretability and Sparse Autoencoders

Recent work in mechanistic interpretability has moved from identifying individual circuits to characterizing the broader feature landscape of transformer models. Sparse autoencoders (SAEs) trained on residual stream activations recover interpretable monosemantic features that are distributed across layers (Cunningham et al., 2023; Templeton et al., 2024). Circuit tracing methods allow attribution of model outputs to specific active features, providing a graph-structured view of which features participate in each computation.

2.2 Reasoning Depth in Language Models

Prior work on reasoning in language models has largely focused on behavioral benchmarks — whether models produce correct answers to multi-step problems (Wei et al., 2022; Kojima et al., 2022). Less work characterizes the internal structure of that reasoning. Chain-of-thought prompting (Wei et al., 2022) improves performance on reasoning tasks but doesn't characterize where in the network the reasoning occurs. Our work takes a different approach: rather than measuring output quality, we measure the depth profile of feature activation for prompts that vary in their reasoning demands while controlling for surface form.

2.3 Layer Specialization in Transformers

There is evidence that transformer layers are not uniform — earlier layers tend to encode syntactic and positional information, while later layers encode more semantic and task-specific content (Tenney et al., 2019; Rogers et al., 2020). If surface recall is more akin to pattern-matching on stored associations, we would expect it to engage later, more directly output-predictive layers. If insight requires integration of distributed knowledge, we would expect it to involve more mid-network processing. Our metric — mean layer of activated SAE features — operationalizes this intuition.

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3. Methods

3.1 Prompt Pair Construction

We construct matched prompt pairs on the following criteria:

1. **Vocabulary match**: Surface and insight prompts for a given topic draw from the same lexical domain
2. **Token count match**: Prompts are matched within ±2 tokens
3. **Syntactic structure match**: Both prompts are sentence-completion fragments of comparable syntactic form
4. **Semantic contrast**: Surface prompts invoke direct factual statements; insight prompts invoke causal or mechanistic explanation of the same phenomenon

Example pair (topic: *digestion*):

- Surface: *"Digestion breaks food into nutrients and the stomach uses acids to process what you"* → target: *eat*
- Insight: *"Digestion converts complex molecules into absorbable units and each stage hands off to the"* → target: *next*

Both prompts concern digestion, use similar vocabulary, and complete at the same syntactic position. The surface prompt invokes a familiar associative completion; the insight prompt requires encoding a mechanistic handoff relationship.

Pairs were constructed for 75 topics across three phases: exploratory (n=30), held-out replication (n=15), and power replication (n=30). Topics were non-overlapping across phases. Pairs were authored by the first author, with later phases using LLM-assisted generation (Claude, Anthropic) followed by manual verification that matching criteria were satisfied and that the semantic contrast was preserved. The target completions were not constrained to match between surface and insight prompts; what is matched is the prompt structure, not the completion.

3.2 Negative Result: Surface Statistics

Prior to circuit tracing, we conducted five rounds of statistical analysis on surface text properties across three model scales (GPT-2 Small 117M, Gemma-2 2B, Mistral 7B), testing whether any combination of the following features separates surface from insight prompts:

- Zipf frequency (per-token average log frequency)
- Flesch-Kincaid readability grade level
- Type-token ratio

- Mean N-gram log probability (unigram, bigram, trigram)
- Syntactic depth (mean parse tree depth)
- TF-IDF classification accuracy

Across all metrics and all pairs, no consistent separation was found that could not be attributed to distributional confounds. When vocabulary controls were strict enough to eliminate TF-IDF signal (accuracy at chance, ~50%), probe classification was not significant (best p = 0.175 at layer 4, Mistral 7B). When constraints were relaxed enough to yield sufficient pairs, TF-IDF accuracy rose above chance (0.667–0.842), invalidating the separation as detecting vocabulary rather than depth. Full results are reported in Appendix A.

3.3 Circuit Tracing and Feature Depth

We use `circuit_tracer` (Lindsey et al., 2025) with Gemma-2-2B (26 layers) loaded via TransformerLens. For each prompt, we compute feature attribution graphs using the `attribute()` function with the following parameters:

- `batch_size=16`
- `max_feature_nodes=500`
- `offload="cpu"`
- `dtype=torch.bfloat16`

For each attribution graph, we extract the `active_features` tensor of shape `(n_active, 3)` where column 0 is the layer index. We compute the mean layer index across all active features as our depth metric:

```
depth = float(np.mean(active_features[:, 0].float().cpu().numpy()))
```

Active feature counts per prompt ranged from approximately 12,000 to 28,000 (see Appendix A.6). The `max_feature_nodes=500` parameter constrains the attribution graph construction step, not the active feature extraction; the mean layer metric is computed over all active features.

The delta for each pair is defined as:

```
delta = surface_mean_layer - insight_mean_layer
```

A positive delta indicates that surface prompts activate features at higher (more output-proximal) layers, consistent with the hypothesis. (Note: the exploratory experiment code

used the reversed convention, delta = insight − surface; all results reported here use the surface − insight convention.)

Pairs with zero active features on either prompt are excluded from analysis (attribution extraction failure). This exclusion criterion is purely technical — it reflects CUDA memory errors during attribution, not directional filtering — and is applied before examining the direction of any pair.

3.4 Exclusion Rates Across Phases

Attribution extraction failures were common in the exploratory phase (28/30 pairs excluded, 93%) due to CUDA out-of-memory errors in the initial pipeline configuration. Between the exploratory and replication phases, memory management was improved (explicit GPU cache clearing between pairs, reduced batch size for problematic prompts), reducing exclusions to 4/15 (27%) in the replication. By the power replication, the pipeline was stable: 0/30 excluded (0%). The improvement reflects engineering fixes to the attribution pipeline, not any change in pair selection or exclusion criteria.

3.5 Pre-registration

Before running each replication, the directional hypothesis (`surface_mean_layer > insight_mean_layer`) was pre-registered. The statistical test (one-tailed paired t-test in the predicted direction) was specified in advance.

3.6 Sensitivity Analysis

For both replications, we ran leave-one-out (LOO) sensitivity analysis: for each pair, we removed it and recomputed the t-test on the remaining n−1 pairs. This tests whether any single pair is responsible for the significant result.

3.7 Robustness Checks

To address potential concerns about the parametric t-test, we also report the nonparametric Wilcoxon signed-rank test for both replications (one-tailed, alternative: delta > 0). We assess normality of the delta distributions with the Shapiro-Wilk test.

Because we test the same directional hypothesis across three samples (exploratory, replication, power), we report Bonferroni-corrected significance thresholds ($\alpha = 0.05/3 \approx 0.017$) alongside the uncorrected values. Note that the exploratory phase was not a formal hypothesis test (n=2 valid), so the correction is conservative.

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4. Results

4.1 Negative Result: Surface Statistics Fail

Across five rounds of analysis on prompt pairs at three model scales, no surface text metric reliably separates surface recall from insight prompts. The key convergent finding: either vocabulary controls are strict enough to bring TF-IDF to chance but the yield of valid pairs collapses (6/70 = 8.6% clean pairs in the strictest condition), or enough pairs survive but TF-IDF remains above chance (0.667–0.842), indicating the “depth” signal is entangled with distributional word statistics.

On the strictest zero-new-word pairs (n=6, Mistral 7B), probe accuracy was not significant at any layer (best: 0.75 at layers 4 and 16, $p = 0.175$). A broader set of 10 vocabulary-controlled pairs on Mistral 7B (TF-IDF at chance, 0.50) showed probe accuracy reaching 0.80–0.90 at mid-to-late layers, but with only 200 permutations the p-values ranged from 0.000 (layer 12) to 0.170 (layer 0). Without eliminating TF-IDF confounds completely, these results cannot be attributed to depth rather than residual distributional signal. See Appendix A.2 for full layer-by-layer results.

This finding is not merely a null result. It is a positive claim: **explanatory depth necessarily manifests in distributional word statistics**. The signal and the confound are the same thing. This has implications for any probing study targeting higher-order cognitive properties (see Discussion, Section 5.4).

4.2 Exploratory Finding: Circuit Depth Separates the Pairs

We submitted 30 exploratory pairs to the attribution pipeline. CUDA memory errors during attribution reduced the yield to 2 valid pairs (fire, inflation), a 93% failure rate reflecting the initial pipeline configuration (see Section 3.4). On these two pairs:

Metric	Value
Surface mean layer	8.785 (fire: 8.725, inflation: 8.845)
Insight mean layer	8.459 (fire: 8.451, inflation: 8.467)
Mean delta (surface – insight)	+0.326
Pairs in predicted direction	2/2 (100%)

While n=2 is insufficient for statistical inference, both pairs showed surface activation concentrated at higher layers than insight activation. The direction was consistent and motivated the pre-registered replication. We do not report p-values for this phase.

4.3 Pre-registered Replication

On the held-out replication sample, 15 pairs were run; 11 yielded valid bilateral attributions. Four pairs were excluded for attribution extraction failure: interest_rates (insight attribution failed), democracy_decay (surface attribution failed), reputation (insight attribution failed), and momentum_social (surface attribution failed). No pair was excluded for directional reasons; failures were symmetric (2 surface-side, 2 insight-side).

Metric	Value
n valid pairs	11
Surface mean layer	8.378 (SD = 0.267)
Insight mean layer	8.260 (SD = 0.195)
Mean delta	+0.118 (SD = 0.192)
% surface deeper	72.7% (8/11)
t(10)	2.04
p-value (one-tailed, pre-registered)	0.035
Wilcoxon signed-rank p (one-tailed)	0.051
Shapiro-Wilk p (normality)	0.381

The parametric test is nominally significant ($p = 0.035$) but the nonparametric Wilcoxon test is marginal ($p = 0.051$). The result does **not** survive Bonferroni correction for three samples (threshold $\alpha = 0.017$). The delta distribution passes Shapiro-Wilk ($p = 0.38$), supporting the parametric test's validity, but the marginal Wilcoxon and failed Bonferroni underscore the fragility at this sample size.

Per-pair deltas ranged from −0.299 (seasons_plants) to +0.342 (stars), with 8 of 11 pairs in the predicted direction. Topics showing the largest effects included stars (+0.342), digestion (+0.336), cancer (+0.260), and electricity (+0.221).

Sensitivity analysis: LOO analysis confirms the fragility. Only 4 of 11 LOO subsets retain $p < 0.05$: photosynthesis (LOO $p = 0.032$), negotiation ($p = 0.028$), propaganda ($p = 0.037$), and seasons_plants ($p = 0.003$). The remaining 7 subsets produce p-values between 0.051 and 0.070. Removing the seasons_plants pair (the largest counter-directional outlier at $\delta = -0.299$) yields the strongest subset (LOO $t = 3.61$, $p = 0.003$), while removing the largest positive outliers weakens the result. The directional finding is stable; the fragility reflects power, not effect inconsistency. This motivated the power replication.

4.4 Power Replication

To address the n=11 fragility, we generated 30 additional held-out pairs on new topics (no overlap with prior 45 pairs), pre-registered the same directional hypothesis, and ran attribution under the improved pipeline (Section 3.4). All 30 pairs yielded valid bilateral attributions (zero exclusions).

Metric	Value
n valid pairs	30
Surface mean layer	8.359 (SD = 0.290)
Insight mean layer	8.102 (SD = 0.250)
Mean delta	+0.257 (SD = 0.313)
% surface deeper	83.3% (25/30)
t(29)	4.50
p-value (one-tailed, pre-registered)	0.000051
Wilcoxon signed-rank p (one-tailed)	0.000055
Cohen's d	0.82
Bonferroni-corrected threshold (α/3)	0.017
Survives Bonferroni?	Yes ($p \ll 0.017$)
Shapiro-Wilk p (normality)	0.894

The effect size (delta = 0.257) is closer to the exploratory estimate (0.326) than to the replication (0.118), consistent with the replication being underpowered rather than the effect being inflated. The delta distribution passes Shapiro-Wilk (p = 0.89), and the parametric and nonparametric tests converge.

Per-pair deltas ranged from −0.452 (nuclear_energy) to +0.850 (trauma). The five largest effects were trauma (+0.850), cryptocurrency (+0.812), black_holes (+0.652), blood_pressure (+0.636), and monopoly (+0.624). Five pairs showed counter-directional effects: nuclear_energy (−0.452), immune_response (−0.156), bacteria_growth (−0.140), fermentation (−0.126), and metabolism (−0.120).

Sensitivity analysis: LOO analysis confirms full robustness. All 30 LOO subsets retain significance at $p < 0.001$ (maximum LOO p = 0.000111, upon removing monopoly). The result

does not depend on any individual pair. Removing the largest counter-directional outlier (nuclear_energy, $\delta = -0.452$) yields the strongest subset (LOO $t = 5.27$, $p = 0.000007$).

4.5 Feature Count Asymmetry: A Confound Analysis

A reviewer might reasonably ask: do insight prompts systematically activate more features, and could this mechanically lower their mean layer index?

The answer to the first question is yes. Across both replications, insight prompts activate significantly more features than surface prompts (replication: 18,962 vs. 15,953 mean features, paired $t = -4.75$, $p < 0.001$; power: 22,456 vs. 16,541, paired $t = -11.82$, $p < 0.001$). Insight prompts activate roughly 1.4× more features on average.

However, the asymmetry does not drive the depth effect:

1. **No correlation**: The feature count ratio (insight/surface) does not predict the delta across pairs ($r = -0.129$, $p = 0.496$ in the power replication).
2. **Median split**: Splitting pairs by feature count ratio, the depth effect holds in both halves — low-asymmetry pairs (n=15): mean delta = +0.310, $p = 0.0002$; high-asymmetry pairs (n=15): mean delta = +0.204, $p = 0.023$.
3. **Direction of bias**: The weak correlation ($r = -0.13$) is *negative* — greater feature count asymmetry is associated with *smaller* deltas, meaning the confound, if anything, works against the reported effect.
4. **Feature count does not predict layer depth**: Insight feature count is not significantly correlated with insight mean layer ($r = -0.162$, $p = 0.392$).

We conclude that the feature count asymmetry is real (insight prompts do engage more features) but orthogonal to the mean-layer depth metric. The higher feature count may reflect the broader scope of causal/mechanistic reasoning, but it does not explain why those features cluster at lower layers.

4.6 Combined Analysis

Pooling the two pre-registered replications (n = 41 valid pairs, no topic overlap), the combined result is: mean delta = +0.219, $t(40) = 4.85$, $p = 0.000010$ (one-tailed).

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5. Discussion

5.1 What This Shows

The combination of the negative result (surface statistics fail) and the positive result (circuit

depth succeeds) tells a specific story: the surface/insight distinction is not a property of the text — it is a property of the computation. Two prompts matched on every measurable surface dimension can produce systematically different activation depth profiles. That difference is recoverable from SAE feature graphs but not from the prompts themselves.

The mechanistic interpretation is consistent with prior work on layer specialization: surface recall likely engages high-confidence stored associations that activate near-output features directly, while insight prompts require integration across concepts — a process that recruits more distributed, earlier-layer representations before converging.

The effect size (Cohen's d = 0.82) is large by convention, though the absolute difference in mean layer index is modest (~0.26 layers on a 26-layer model, or ~1% of total depth). This suggests the metric captures a real but subtle computational signature — the kind of signal that would be invisible to behavioral evaluation but meaningful at the circuit level.

5.2 What This Doesn't Show

This is not a general reasoning scorer. The finding holds on carefully matched pairs across a 2B parameter model. We have not tested:

- Whether the effect generalizes to larger models (7B, 70B+)
- Whether it holds on uncontrolled naturalistic text
- Whether the metric is robust to prompt reformulation of the same semantic content
- Whether the effect is architecture-specific to Gemma-2 or present across model families
- Whether the absolute layer values or only the relative ordering is meaningful
- Whether the effect correlates with next-token prediction confidence (ruling out prediction uncertainty as the sole driver)
- Whether activation-strength-weighted mean layer yields consistent results

These are tractable questions with the same methodology and represent the natural next steps.

5.3 Limitations

Several limitations warrant explicit discussion:

Small model. Gemma-2-2B has only 26 layers. A mean layer shift of ~0.26 is small in absolute terms. Whether the effect scales, reverses, or vanishes in larger models is unknown.

Matched pairs only. Our prompts are tightly controlled for vocabulary, token count, and syntax. Whether the effect holds on naturalistic, uncontrolled text — where surface statistics differ — is an open question. The effect may be specific to the matched-pair regime where surface confounds are eliminated.

Single architecture. All circuit tracing was done on Gemma-2. Different model families with different SAE decompositions may yield different depth profiles.

Prediction uncertainty confound. The depth effect could in principle correlate with prediction difficulty rather than reasoning depth per se. If insight prompt completions are simply harder for the model to predict (lower next-token probability), the deeper feature activation could reflect uncertainty processing rather than causal reasoning. We did not measure model confidence on target completions, so this alternative cannot be ruled out with the current data. Future work should report next-token log-probabilities for both conditions.

Exploratory phase was severely underpowered. Only 2 of 30 pairs yielded valid attributions, making the exploratory estimate (delta = +0.326) unreliable despite being directionally correct. The replication and power studies are the evidentiary core.

Pre-registration method. The directional hypothesis was pre-registered in local JSON files with timestamp metadata, not on a formal pre-registration platform (e.g., OSF, AsPredicted). The pre-registration is verifiable through version-controlled files and commit history, but does not carry the same assurance as platform-based pre-registration. We will deposit the full experimental record on a public repository prior to submission.

One metric. Mean layer index is a simple summary statistic. Richer characterizations — layer variance, feature co-activation patterns, layer-specific feature density — may reveal more nuanced differences.

5.4 The Frequency Paper Connection

This work complements a prior result (darkflobi, 2026) showing that surface text statistics including word frequency distributions cannot distinguish surface from insight text. Across five iterations at three model scales (GPT-2 Small 117M, Gemma-2 2B, Mistral 7B), that work established that the lexical fingerprint is constitutive of the distinction — expressing explanatory depth necessarily alters distributional word statistics. The present paper extends this by establishing a positive signal on the computation side.

Together they tell a complete story: you cannot tell from the text whether a model is recalling or reasoning, but you can tell from the circuit.

5.5 Methodological Contribution

The primary contribution may be the methodology rather than the specific finding. The matched-pair construction protocol — controlling vocabulary, token count, and syntactic structure while varying semantic depth — provides a clean test bed for any hypothesis about reasoning depth in language models. The use of SAE feature mean layer as a depth metric is simple, interpretable, and model-agnostic (any model with available SAEs can be tested).

5.6 Counter-Directional Pairs

Five of 30 power replication pairs showed effects opposite to the predicted direction (insight deeper than surface). The strongest was nuclear_energy ($\delta = -0.452$). Four of the five counter-directional pairs involve biological or chemical processes (immune_response, bacteria_growth, fermentation, metabolism) where "surface" prompts may themselves invoke complex process knowledge, blurring the surface/insight boundary for these specific topics. Investigating what distinguishes concordant from discordant pairs is a natural follow-up — it may reveal boundary conditions of the effect or properties of prompts that confound the surface/insight distinction.

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6. Conclusion

Surface statistics fail. Circuit structure succeeds. The distinction between surface recall and explanatory reasoning in language models is not recoverable from input text properties but is recoverable from the depth profile of sparse autoencoder feature activation. The core evidence comes from a pre-registered power replication: n=30 held-out pairs, $t(29) = 4.50$, $p = 0.000051$, Cohen's $d = 0.82$, robust to nonparametric test (Wilcoxon $p = 0.000055$), leave-one-out analysis (all subsets $p < 0.001$), Bonferroni correction, and feature-count confound control. An earlier replication at n=11 is directionally consistent but statistically fragile, underscoring the importance of adequate sample size for this effect.

Generalization to larger models, additional architectures, and naturalistic text is the immediate next step.

Appendix A: Surface Statistics Results

A.1 Experiment Progression

Experiment	Model	Pairs	TF-IDF Accuracy	Best Probe p	Outcome

01	GPT-2 Small (117M)	8	Not measured	—	Weak signal (cosine dist. 0.004), confounded by vocabulary
04	GPT-2 Small (117M)	8	~0.50 (chance)	p = 0.99	Not significant; vocabulary controls working
05	Gemma-2 (2B)	8	~0.50 (chance)	p = 0.948	Not significant at larger scale
10A	Mistral (7B)	95	0.842	p ≈ 0.000 (all layers)	100% probe accuracy — detecting vocabulary, not depth
10B	Mistral (7B)	6 clean (of 70)	0.667	p = 0.175 (layer 4)	Clean pairs: not significant, but TF-IDF still > chance

Note: Experiment 10A and 10B represent the same dataset under different filtering. 10A used all 95 hand-authored pairs without vocabulary control; 10B applied a zero-new-word constraint and retained only 6 pairs where no word appeared exclusively in one condition. Even under this strictest control, TF-IDF accuracy remained above chance (0.667), indicating that distributional shifts among shared vocabulary carry enough signal for bag-of-words separation.

A.2 Mistral 7B Layer-by-Layer Probe Results (Vocabulary-Controlled, n=10 pairs, TF-IDF=0.50)

A separate set of 10 vocabulary-controlled pairs (TF-IDF at chance) was probed across all Mistral 7B layers:

Layer	Probe Accuracy	Null Mean	p-value (200 permutations)
0	0.60	0.495	0.170
4	0.75	0.499	0.030
8	0.70	0.503	0.045
12	0.80	0.503	< 0.005
16	0.90	0.502	0.005

20	0.80	0.504	0.005
24	0.80	0.503	0.010
28	0.80	0.502	0.015
31	0.70	0.500	0.050

Probe accuracy peaks at mid layers (12–16) and decreases at output layers, consistent with the hypothesis that the distinction is encoded at intermediate representations. However, these results are from a separate probe (PCA + linear classifier on residual stream activations) and a different model (Mistral 7B), not the SAE circuit tracing method used for the main results. They are included as supporting context, not as evidence for the primary claim.

A.3 Replication Per-Pair Data (Experiment 12, n=11 valid of 15)

Topic	Surface Layer	Insight Layer	Delta	Direction
Digestion	8.226	7.891	+0.336	✓ Predicted
Photosynthesis	8.180	8.202	−0.021	✗ Counter
Negotiation	8.245	8.290	−0.045	✗ Counter
Cancer	8.739	8.479	+0.260	✓ Predicted
Music	8.342	8.213	+0.128	✓ Predicted
Electricity	8.710	8.489	+0.221	✓ Predicted
Propaganda	8.429	8.410	+0.019	✓ Predicted
Seasons/Plants	7.809	8.109	−0.299	✗ Counter
Addiction Cycle	8.585	8.434	+0.151	✓ Predicted
Stars	8.363	8.021	+0.342	✓ Predicted
Immune Memory	8.528	8.324	+0.203	✓ Predicted

Excluded pairs (attribution failure on one side): interest_rates (insight failed), democracy_decay (surface failed), reputation (insight failed), momentum_social (surface failed).

A.4 Power Replication Per-Pair Data (Experiment 12 Power, n=30 valid of 30)

Topic	Surface Layer	Insight Layer	Delta	Direction
Trauma	8.619	7.769	+0.850	✓ Predicted
Cryptocurrency	8.683	7.871	+0.812	✓ Predicted
Black Holes	8.575	7.922	+0.652	✓ Predicted
Blood Pressure	8.522	7.886	+0.636	✓ Predicted
Monopoly	8.037	7.413	+0.624	✓ Predicted
Earthquakes	8.534	7.918	+0.616	✓ Predicted
Coral Reefs	9.035	8.506	+0.528	✓ Predicted
Dreams	8.582	8.130	+0.451	✓ Predicted
Social Norms	8.643	8.213	+0.430	✓ Predicted
Enzyme	8.657	8.240	+0.418	✓ Predicted
Obesity	8.357	7.979	+0.378	✓ Predicted
Empathy	8.387	8.039	+0.348	✓ Predicted
Hurricanes	8.437	8.100	+0.337	✓ Predicted
Language Acquisition	8.484	8.198	+0.286	✓ Predicted
Ocean Currents	8.268	8.010	+0.258	✓ Predicted
Stock Bubbles	8.611	8.396	+0.215	✓ Predicted
Supply Chains	8.294	8.079	+0.215	✓ Predicted
Bird Migration	8.532	8.327	+0.205	✓ Predicted
Opioid Dependence	8.242	8.074	+0.167	✓ Predicted
Ocean Acidification	8.378	8.253	+0.124	✓ Predicted
Drug Resistance	8.357	8.317	+0.040	✓ Predicted
Leadership	8.331	8.291	+0.040	✓ Predicted

Plate Tectonics	7.752	7.724	+0.027	✓ Predicted
Volcanoes	8.117	8.097	+0.020	✓ Predicted
Taxation	8.293	8.277	+0.015	✓ Predicted
Metabolism	8.223	8.343	−0.120	× Counter
Fermentation	7.787	7.912	−0.126	× Counter
Bacteria Growth	8.134	8.275	−0.140	× Counter
Immune Response	7.751	7.907	−0.156	× Counter
Nuclear Energy	8.155	8.607	−0.452	× Counter

A.5 Leave-One-Out Sensitivity: Replication (n=11)

Pair Removed	LOO t	LOO p (one-tailed)	Significant at α=0.05?
Seasons/Plants	3.606	0.003	✓
Negotiation	2.185	0.028	✓
Photosynthesis	2.121	0.032	✓
Propaganda	2.026	0.037	✓
Music	1.825	0.051	×
Addiction Cycle	1.792	0.053	×
Immune Memory	1.726	0.059	×
Electricity	1.707	0.061	×
Cancer	1.670	0.065	×
Stars	1.617	0.070	×
Digestion	1.620	0.070	×

A.6 Feature Count Summary

Phase	Surface Features (mean ± SD)	Insight Features (mean ± SD)	Ratio	Paired t	p

Replication (n=11)	15,953 ± 1,437	18,962 ± 2,115	1.19×	−4.75	< 0.001
Power (n=30)	16,541 ± 2,099	22,456 ± 3,055	1.36×	−11.82	< 0.001

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